

CONFIDENT IN AI OR MYSELF? COMPUTATIONAL MODEL OF DUAL-CONFIDENCE DYNAMICS IN AI-ASSISTED DECISION-MAKING

Manxin Chen¹, Leah Chong^{1,*}

¹The University of Texas at Austin, Austin, TX

ABSTRACT

In AI-assisted engineering design workflows, designers often repeatedly evaluate AI suggestions and decide whether to accept, reject, or modify them. These reliance decisions are shaped by two evolving beliefs: confidence in the AI and self-confidence. The trajectories of these beliefs are modeled in Chong et al.'s dual-channel model that updates both confidences trial-by-trial, setting a foundation for trust calibration in human-AI collaboration. However, the model's assumptions, such as binary reliance decision, lead to limited accuracy in the self-confidence channel, as well as limited applicability to broader AI-assisted design scenarios. Therefore, this work reformulates the model with four changes: 1) a modular update skeleton that introduces momentum into confidence updating, 2) an expanded experience encoding that represents modification as partial credit, 3) an accumulated expectation mechanism driven by experience rather than confidence history, and 4) asymmetric slow-learning forms in the two channels. Evaluated on an empirical human data from an AI-assisted chess task, the reformulation improves the model fit from adjusted $R^2 = 0.753$ to 0.792 for the confidence in AI channel and $R^2 = 0.424$ to 0.632 for the self-confidence channel. Our reformulated model paves a pathway from latent confidence dynamics to confidence detection and subsequent adaptive interventions in human-AI design workflows.

Keywords: Human-AI collaboration, AI-assisted decision-making, confidence, computational modeling, trust calibration

NOMENCLATURE

Baseline model parameters

α_e	Rate factor for experience term
α_a	Rate factor for accumulated confidence term
α_b	Rate factor for bias term
γ	Recency parameter in baseline accumulator
ω_i	Subjective weight for experience type i

Reformulated model parameters

α_{fast}^k	Fast learning rate for channel k
α_{slow}^k	Slow learning rate for channel k
α_a^k	Expectation learning rate for channel k
β^k	Momentum coefficient
g_{align}	Alignment amplification coefficient
ρ	Weight assignment parameter for modify trials

State variables

C_n^k	Confidence belief for channel k at trial n , $C_n^k \in [0, 1]$
A_n^k	Accumulated expectation state for channel k at trial n
v_n^k	Velocity (change state) for channel k at trial n
E_n	Experience signal at trial n
F_n^k	Net driving signal for channel k at trial n
F_{fast}^k	Driving signal component for fast learning
F_{slow}^k	Driving signal component for slow learning
$e_{i,n}$	Binary experience indicator for type i , $e_i \in \{0, 1\}$
Align_n	Alignment flag at trial n , $\text{Align}_n \in \{0, 1\}$
B	Bias term, set to initial confidence C_0

Indices and sets

k	Channel index, $k \in \{\text{self}, \text{AI}\}$
n	Trial index
i	Experience type index, $i \in \{1, 2, 3, 4\}$

1. INTRODUCTION

Advances in the capability of artificial intelligence (AI) are bringing about new human-AI interaction modes in the engineering design process, largely shifting from discrete delegation of tasks to iterative collaboration. In AI-assisted workflows, AI systems often generate suggestions that designers must assess and decide whether to accept, reject or modify [1]. An example is the generative computer-aided design (CAD) workflow: designers receive AI-generated design alternatives in rapid succession, compressing traditionally multi-step processes (e.g. ideation and verification) into a single evaluation [2, 3]. Consequently, engineers encounter more design variations and iterate more and often faster than in traditional workflows. This rapid, repeated

*Corresponding author: leah.chong@austin.utexas.edu

decision-making extends beyond generative CAD and is common in many AI-assisted engineering design scenarios.

In this new context of repeated human-AI interaction (HAI), appropriate reliance decisions (e.g., accept, reject, and modify) are crucial for successful collaboration. Under-reliance is costly; ignoring helpful AI suggestions means missing high-performing alternatives and spending extra time and resources exploring designs the AI could have surfaced early [4]. Over-reliance can also be expensive and even hazardous; flawed designs can carry downstream consequences for analysis, prototyping, manufacturing, and deployment. For example, if an engineer accepts an AI-generated topology that overlooks latent failure modes, the result may be late-stage redesign, scrapped prototypes, or unsafe field performance [5]. Therefore, it is important that designers' perceived capability of an AI system aligns with the system's actual performance, a concept that is coined as trust calibration [6, 7].

To support trust calibration, many human-computer interaction (HCI) researchers, including those specifically in HAI, have explored interface mechanisms to communicate model reliability to users, such as uncertainty visualizations and calibrated confidence disclosures [8–10]. While these mechanisms can improve reliability judgments, they often increase cognitive load particularly in time-sensitive settings [9, 11, 12]. High cognitive load pushes users toward simplified heuristics that leads to ineffective decision-making [13, 14], leaving the mis-reliance issue unresolved. An alternative approach uses adaptive interventions that prompt users when they appear to over-rely or under-rely on AI. Examples include surfacing explanations, slowing interactions, and tailored support to the user's current trust state [15–19]. While promising, these methods often treat trust as an unobserved latent state and require real-time detection of user trust, which is a capability that is not fully realized yet. Realizing such real-time prediction requires understanding how trust in AI evolves over repeated interactions. Yet existing works in this domain offer limited characterization of trust dynamics: whether changes are gradual or abrupt, symmetric or asymmetric to positive and negative feedback, or dominated by recent trials or longer history [20, 21]. A clearer temporal description is needed before real-time trust prediction becomes feasible.

To address this gap, Chong et al. (2022) proposed a computational model of two constructs that shape AI trust and reliance decisions over time [22]: confidence in AI and self-confidence. While these constructs do not capture the full breadth and complexity of trust, they align with two major antecedents emphasized in prior trust research: perceived ability of the trustee and the trustor's propensity or willingness to rely [23]. Confidence in AI reflects how capable users think the AI is for a given task, while self-confidence moderates willingness to accept external assistance. Understanding how these constructs evolve across interactions can inform real-time trust prediction and adaptive intervention design.

Although Chong et al.'s model provides a useful foundation [22], two limitations constrain its applicability to broader engineering design contexts and beyond. First, the model treats reliance decisions as binary (accept/reject), neglecting more complex behaviors such as partial adoption and modification. Second, it shows lower predictive accuracy for self-confidence, suggesting

that the self-confidence update mechanism remains insufficiently captured. To address these limitations, we reformulate the model by implementing the following four changes:

1. an modular update skeleton that introduces momentum into confidence updating,
2. an expanded experience encoding that captures modification decision,
3. an accumulated expectation mechanism driven by prediction error, and
4. an asymmetric parameterization of slow learning between self-confidence and confidence in AI channels with alignment-sensitive amplification.

We then evaluate whether and how this reformulation improves the model's ability to reproduce empirical confidence trajectories for both constructs with an AI-assisted chess task dataset [24]. While the chess task is not a design task, this dataset is chosen for its availability and resemblance to various AI-assisted design tasks. The dataset includes human-AI interaction data, such as AI suggestions, reliance decisions, and feedback, over multiple trials, which can be found in iterative AI-assisted design tasks. Furthermore, this dataset is used to evaluate Chong et al.'s model [22], providing a good reference for comparing the performance of the reformulated model. The evaluation results show that the reformulation substantially improves predictive accuracy, especially for the self-confidence channel.

2. RELATED WORK

2.1. Human-AI Collaboration in Engineering Design

Engineering design increasingly involves iterative human-AI collaboration, where AI acts as a generative partner rather than a back-end optimizer. A framework for human-AI teaming in engineering design distinguishes multiple integration modalities, from passive tools invoked on demand to proactive partners and process managers that guide workflow [25]. As AI moves beyond its role as a tool, AI-assisted design often involves AI proposing candidate artifacts or design directions, then designers critiquing, steering, and revising them over multiple cycles.

Various empirical studies illustrate the impact of this shift on teamwork and design process and outcomes. In recent drone design experiments, hybrid teams with AI agents showed improved coordination, broader exploration, and stronger adaptation to disruptions than human-only teams [1], and an AI process manager matched human managers in shaping team problem-solving behavior [26]. However, adding a high-performing AI partner does not consistently improve design outcomes. Zhang et al. (2021) found that high-performing teams were hindered while low-performing teams initially benefited by a high-performing AI teammate, indicating that AI effects depend on the existing team dynamic [27]. When AI agents are proactive teammates, the collaboration sometimes depends more on how humans adapt their communication than on the AI's capabilities. [28].

These findings suggest that design outcomes are jointly produced by human and AI contributions, making credit and blame difficult to assign to a single actor. In fact, human designers

tend to blame themselves even for negative outcomes caused by their AI partner [29]. This attribution ambiguity is a defining feature of collaborative design workflows: the same outcome may be attributed to AI capability, human skill, or their interaction, leading to different updates in beliefs about the AI and the self [4, 30, 31]. Prior work supports this dual-belief view, showing that confidence in AI responded primarily to poor AI performance, whereas self-confidence was sensitive to changes in AI accuracy rather than to the level of accuracy itself [29]. These findings motivate modeling the beliefs in self and AI in parallel to accurately understand credit/blame attribution patterns in AI-assisted decision-making and the associated design outcomes.

2.2. Trust Calibration in AI-Assisted Decision Making

Trust calibration, defined as aligning reliance with actual system capability, is central to preventing misuse (over-reliance) and disuse (under-reliance) of automation [32]. Most calibration research has focused on providing users with information to support their judgment of AI advice, such as uncertainty displays, confidence scores, or other explanations. Such cues can shift reliance behavior, though their effects vary with task context and user goals [7, 33]. Part of this variability may also stem from how users interpret evidence, penalizing algorithms more heavily after errors than rewarding them after successes, contributing to algorithm aversion [34]. Calibration therefore depends not only on what evidence is presented, but also on how that evidence is processed over time.

Another popular approach to calibrating trust is behavioral interventions, detecting when a user appears to over-rely or under-rely on AI and actively adjusting the interaction. Some examples include requiring independent justification, slowing acceptance, or modulating support intensity [15, 16, 18, 19]. These interventions require real-time inference of the user's latent trust state from observable behavior. However, this inference has not been sufficiently realized, given the difficulty of capturing the dynamic and multi-faceted trust updating mechanism.

In engineering design, this real-time trust detection is especially challenging. Much of the existing calibration literature assumes that evidence about system performance is relatively interpretable, so that users can judge whether the success or failure should be credited to AI. In design workflows, however, outcomes are often jointly produced through suggestion, critique, and revision, weakening the applicability of this assumption. Trust calibration in such settings depends not only on designers' judgments of AI competence but also on their perceptions of their own contribution to the joint outcome, requiring models that track both of these beliefs over time.

2.3. Modeling Trust Dynamics

Computational modeling efforts of trust and confidence can be organized along two axes: temporal resolution (static vs. dynamic) and belief structure (single-belief vs. multi-belief) [32, 35–37]. Static approaches treat trust as a stable attitude assessed via surveys or post-hoc judgments. They tend to be limited in explaining trial-to-trial fluctuations in repeated human-AI interactions [38]. In contrast, dynamic models commonly represent trust as a latent state updated through experience. Xu

and Dudek (2015) used online Bayesian inference to estimate a single trust variable from observed behavior in human-robot collaboration [39]. Hu et al. (2019) proposed an error-correction formulation in which trust in automation updates trial-by-trial as a function of experience, cumulative trust, and expectation bias [40]. While these models can capture temporal dynamics of trust, they often simplify trust as a single-belief system. However, a single-belief approach likely cannot accurately predict trust and subsequent reliance decisions, as they depend on both perceived system (i.e., AI) capability and self confidence [23, 41].

Subsequent works introduced multi-belief models of trust. Chong et al. (2022) extended Hu et al.'s error-correction model to two channels, confidence in AI and self-confidence, updated in parallel with channel-specific parameters [22, 40]. Williams et al. (2023) modeled the same two beliefs as coupled states in a partially observable Markov decision process, capturing their probabilistic relationship with reliance [37]. Both models establish that reliance is shaped jointly by confidence in AI and self-confidence. Williams et al.'s discrete state categories (high/low) limit continuous trajectory prediction, while Chong et al.'s formulation enables trial-by-trial prediction under a simpler reliance structure. Therefore, this work builds on Chong et al.'s dual-channel model, introducing new mechanisms to improve its alignment with findings from cognitive science and its applicability to AI-assisted design workflows.

3. BASELINE MODEL

In this section, we formalize Chong et al.'s dual-channel confidence model as a baseline for trial-by-trial confidence updating [22]. This model, as shown in Fig. 1, serves as the foundation for subsequent reformulations. This baseline model builds on the dynamic systems perspective on trial-by-trial trust updating introduced by Hu et al. [40]. Given the previous trial's confidence report and the experience signal, the model predicts updated confidence in the current trial.

Per trial n , channel $k \in \{\text{self}, \text{AI}\}$:

Confidence update:

$$C_{n+1}^k = C_n^k + \alpha_e [E_n - C_n^k] + \alpha_a [A_n^k - C_n^k] + \alpha_b [B - C_n^k]$$

Experience:

$$E_n = \sum_{i=1}^4 \omega_i e_{i,n}$$

Accumulated confidence:

$$A_{n+1}^k = \gamma C_n^k + (1 - \gamma) A_n^k, \quad \gamma \in [0, 1], \quad A_0^k = C_0^k$$

Bias:

$$B = C_0^k$$

where

$$C_n^k, E_n, A_n^k, B \in [0, 1], \quad \alpha_e, \alpha_a, \alpha_b, \omega_i \in [0, 1], \quad e_i \in \{0, 1\}$$

FIGURE 1: Chong et al.'s dual-channel confidence model [22]

3.1. Observables and State

The baseline model is positioned for AI-assisted decision-making scenarios with repeated trials, each involving a reliance decision, an outcome, and a confidence report. The model's state is human confidence $C_n^k \in [0, 1]$ over two channels $k \in \{\text{self}, \text{AI}\}$. Each trial yields three observables: reported confidence, reliance decision, and outcome feedback. The latter two combine into experience type $e_{i,n} \in \{0, 1\}$. Chong et al. characterize the observables as follows: (i) confidence reports are the self-report of the latent belief state; (ii) reliance decisions are binary $\in \{\text{accept}, \text{reject}\}$; and (iii) outcome feedback is binary $\in \{\text{positive}, \text{negative}\}$.

3.2. Update Mechanism and its Key Drivers

As demonstrated in Fig. 1, confidence updates trial-by-trial via an error-correction rule that combines three drivers: (i) an experience signal E_n derived from the current trial's reliance decision and feedback, (ii) an exponentially discounted history of past confidence A_n , and (iii) a constant bias B set by initial confidence. Here, error-correction means that confidence is corrected in proportion to the discrepancy (i.e., error) between incoming driving signals and the current confidence level. All three signals are defined on the confidence scale $[0, 1]$, and their associated rate factors $\alpha_e, \alpha_a, \alpha_b \in [0, 1]$ govern how strongly each driver influences each update. The model uses the same functional form for both channels (self vs. AI) with channel-specific parameters.

3.2.1. Experience E_n The experience term encodes the interaction outcome in a given trial as a function of binary reliance decision and binary feedback. Each trial falls into one of the four mutually exclusive experience types $e_{i,n} \in \{0, 1\}$ with $i \in \{1, 2, 3, 4\}$, as shown in Table 1. E_n is then computed as their weighted sum $E_n = \sum_{i=1}^4 \omega_i e_{i,n}$, where the weights $\omega_i \in [0, 1]$ represent the strength of impact of each experience type. Intuitively, E_n provides the outcome-driven "teaching signal" for the update.

TABLE 1: Four experience types defined by reliance decision-feedback combinations

	Accept	Reject
Positive feedback	e_1	e_2
Negative feedback	e_3	e_4

3.2.2. Accumulated Confidence A_n To capture the persistence of confidence beyond the single-trial updates, the model includes an exponentially discounted accumulator of past confidence signals. The recency parameter $\gamma \in [0, 1]$ controls the tradeoff between the most recent confidence C_n and the history of signals A_n : larger γ yields stronger weighting to the recent confidence, while smaller γ yields stronger weighting to the accumulation of past signals.

3.2.3. Bias B The bias term represents a user's general prior predisposition towards AI systems, rather than an evaluation of the specific AI during the trials. The baseline model keeps this term constant for simplicity and initializes it to user's starting confidence [22].

3.3. Model Fitting

We confirm the performance of the baseline model by replicating the model fitting result presented by Chong et al. [22]. The model is fitted to an empirical human dataset, which was collected during a multiple-trial AI-assisted decision-making chess task [24]. The data include 100 participants' experiences and confidence reports through 30 trials (plus three practice trials which are used to establish initial confidence and excluded from all analyses). In each trial, each participant received a move suggestion from a Stockfish¹-powered AI assistant, made a reliance decision, and received outcome feedback (+5 or -5 points). Then, after each trial, they reported their confidence in the AI and self-confidence on a five-point Likert scale coded as $\{0, 0.25, 0.5, 0.75, 1\}$.

For each confidence channel, the optimal model parameters $\{\alpha_e, \alpha_a, \alpha_b, \omega_1, \omega_2, \omega_3, \omega_4, \gamma\}$ are estimated independently for each channel by minimizing least-squares error between predicted and reported confidence over all test trials. The resulting nonlinear regression is solved using a trust-region reflective algorithm with multiple random initializations to assess sensitivity [42]. For both channels, we verify that fitted parameters and the accuracy of the baseline model are consistent with those reported by Chong et al. [22]. The accuracies are adjusted $R^2 \approx 0.753$ for the confidence in AI channel and adjusted $R^2 \approx 0.424$ for the self-confidence channel.

3.4. Baseline Limitations

The performance gap between the two channels suggests that there may be dynamics unique to the self-confidence channel that are not captured by the baseline's outcome-driven error correction mechanism. In addition, two assumptions made by the baseline model motivate our reformulation.

First, reliance decision is encoded as binary (i.e., accept or reject), and therefore cannot represent intermediate behaviors such as modification. In the present dataset, modification is operationalized as cases where the participant's final move differs from both their initial choice and the AI suggestion; although infrequent here, this type of reliance decision becomes the dominant interaction mode in settings such as engineering design workflows where users routinely revise and selectively adopt portions of AI suggestions [30, 43]. The binary encoding collapses an operationally distinct subset of non-accept trials into the reject category.

Second, once the model is fitted, the rate factors α_x are fixed across trials, preventing transient, history-dependent changes in sensitivity. Empirical evidence from reinforcement learning research shows that sensitivity to feedback can shift over time; for example, people tend to be more sensitive immediately after surprising outcomes and less sensitive after a stable sequence [44]. Fixed rate factors cannot capture such context-dependent updating. Therefore, a model architecture that can accommodate time-varying sensitivity provides a more general foundation for future extensions to richer task environments.

¹<https://github.com/official-stockfish/Stockfish>

4. REFORMULATED MODEL

Building on Chong et al.'s dual-channel baseline, we propose a reformulation which includes an initial change to the update skeleton, followed by three independent changes to the driving terms.

1. **Physics-inspired update skeleton** reformulates confidence updating into a modular state-update structure that introduces momentum into confidence updating, providing a scaffold for the following changes.
2. **Expanded experience encoding** captures modification reliance decisions with partial weight assignment.
3. **Expectation update** re-characterizes the accumulated confidence as an accumulated expectation.
4. **Dual-rate learning** allows slow component to take two separate forms for confidence in AI and self channels, with alignment-dependent amplification in the fast component.

Figure 2 shows the summary of the model reformulation. The following subsections detail each change.

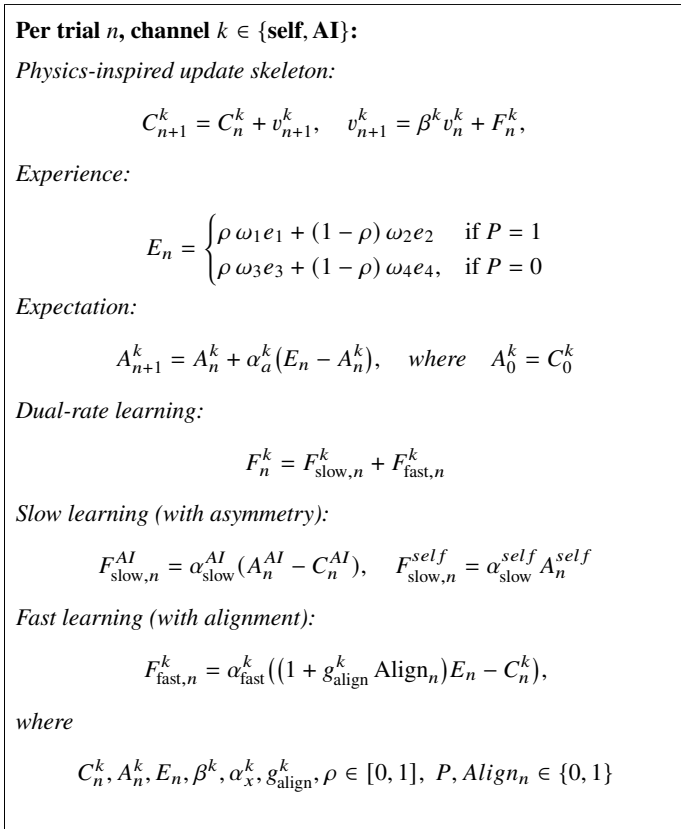


FIGURE 2: Summary of model reformulation

4.1. Physics-inspired Update Skeleton

In the baseline model, confidence is updated based on the current trial's driving terms (experience, accumulated confidence,

and bias) relative to the current confidence state. The accumulated confidence term carries over past confidence states trial-to-trial, supporting state dependence. However, the model does not contain any mechanism that explicitly carries information about the rate of change itself. A large increase or decrease in confidence from trial n to $n + 1$ affects the next update only through the resulting confidence level C_{n+1}^k , failing to capture the momentum of confidence change. Prior work on adaptive learning suggests that learning sensitivity can persist across trials rather than resetting after each event [44–46]. Therefore, the new formulation enables this momentum or continuity in the change process.

To represent this missing momentum explicitly, the reformulation introduces two physics-inspired variables: a net driving signal F_n^k , which aggregates how strongly and in which direction the driving factors push confidence at trial n , and a velocity term v_n^k , which carries recent directional change from one trial to the next. Confidence updates as:

$$C_{n+1}^k = C_n^k + v_{n+1}^k$$

where velocity evolves as:

$$v_{n+1}^k = \beta^k v_n^k + F_n^k$$

The coefficient $\beta^k \in [0, 1]$ controls the momentum; when $\beta^k = 0$, velocity resets each trial and the update mechanism reduces to the baseline's single-step correction, and when $\beta^k > 0$, recent directional change carries over and can influence subsequent updates. The skeleton thus nests the baseline while enabling momentum in change.

With this reformulation, the baseline bias term B no longer appears as a separate additive term. Instead, its role is absorbed into the accumulated term A_n^k , which carries forward both experienced history and prior disposition. By setting $A_0^k = C_0^k$, the role of B is absorbed into the state.

4.2. Expanded Experience Encoding

The baseline encodes each trial's reliance decision as binary (i.e., accept or reject). This encoding cannot represent modification, even though this is an empirically common reliance decision and experience in AI-assisted workflows [43]. The experience signal E_n is computed as a piecewise function of outcome feedback $P \in \{0, 1\}$, where 0 means negative feedback and 1 means positive feedback:

$$E_n = \begin{cases} \rho \omega_1 e_1 + (1 - \rho) \omega_2 e_2 & \text{if } P = 1 \\ \rho \omega_3 e_3 + (1 - \rho) \omega_4 e_4, & \text{if } P = 0 \end{cases}$$

For each feedback condition, modification is represented as shared credit between accept and reject experience types, controlled by $\rho \in [0, 1]$ as shown in Table 2.

4.3. Expectation Update

In the baseline, A_n^k is the exponentially weighted trace of past confidence states: in short, a memory of confidence. In the reformulation, A_n^k is redefined as an accumulated expectation of the experience signal E_n , representing what past trials suggest the next trial is likely to yield. We replace the baseline's accumulator

TABLE 2: Partial weight assignment for modification experience encoding

Positive Feedback ($P=1$)		Negative Feedback ($P=0$)	
Accept (e_1)	Reject (e_2)	Accept (e_3)	Reject (e_4)
ρ	$1 - \rho$	0	0
0	0	ρ	$1 - \rho$

with a prediction-error update inspired by Rescorla-Wagner [47, 48]:

$$A_{n+1}^k = A_n^k + \alpha_a^k (E_n - A_n^k)$$

where $E_n - A_n^k$ is the prediction error. A_{n+1}^k does not change when experience E_n matches expectation A_n^k and adjusts in proportion to the error otherwise. α_a^k controls adaptation speed.

Because A_n^k and C_n^k now track different quantities, their values can diverge to contribute to the update distinctively. For instance, two users may both report $C_n^k = 0.5$, while one may have arrived there after a run of positive trials and the other after a negative run. In the reformulated model, these different recent histories can produce different values of A_n^k .

4.4. Dual-rate Learning

Dual-rate models in sensorimotor adaptation research proposed that motor learning has two components: a fast process that corrects large errors quickly and a slow process that retains learning over longer timescales [49, 50]. We implement a similar decomposition for confidence updating as it enables channel-specific and rate-specific parameterizations in the model. The net driving signal is decomposed into a fast component that reacts to the current trial's experience and a slow component that reflects what has been learned over multiple trials:

$$F_n^k = F_{\text{slow},n}^k + F_{\text{fast},n}^k$$

4.4.1. Slow Component (Channel-Specific) Confidence in AI and self-confidence channels are given structurally different slow learning form. The motivation for this channel-specific slow learning comes from one key empirical finding: Möbius et al. found that people update beliefs about themselves more conservatively and asymmetrically than beliefs about an external agent [51]. This suggests that self-evaluation and evaluation of an external system may follow different updating dynamics.

For confidence in AI, accumulated expectation can become a reference against which new confidence signal is compared [52]. This error-correction form is as follow:

$$F_{\text{slow},n}^{AI} = \alpha_{\text{slow}}^{AI} (A_n^{AI} - C_n^{AI})$$

Here, accumulated expectation A_n^{AI} acts as a reference point: the user's running estimate of how capable the AI is. When confidence overshoots or undershoots relative to accumulated expectation, the error-corrective coupling brings it back. This mechanism corresponds to what Rumelhart and Norman refer to as *tuning*: parametric adjustment of an existing belief toward a stable target [52].

For self-confidence, evidence about one's own performance accumulates without a specific target value [52, 53], taking a different form:

$$F_{\text{slow},n}^{\text{self}} = \alpha_{\text{slow}}^{\text{self}} A_n^{\text{self}}$$

Rather than pulling confidence toward a reference, A_n^{self} contributes directly as a signal that shifts confidence. This structure more closely resembles *accretion*: the gradual accumulation of information that shifts a representation without a discrete comparison process [52, 53].

A notable property of the accretive form is that $F_{\text{slow}}^{\text{self}}$ is non-negative, producing structural resistance to sustained downward revision. Self-confidence can still decline when the fast component dominates, but it must overcome a persistent positive drive, whereas confidence in AI faces a symmetric pull toward a reference. This is consistent with two well-documented patterns in the psychology literature. Möbius et al. found that people discount negative feedback more heavily when updating beliefs about themselves than when performing identical inference about an external agent [51]. Bandura's account of self-efficacy describes a similar resistance: once self-efficacy beliefs are established, they tend to persist even under contradictory evidence [54, 55].

4.4.2. Fast Component (Shared Across Channels) Beyond the reliance decision and outcome feedback encoded in E_n , one additional observable can influence how strongly experience drives confidence: whether the user's own idea aligned with the AI suggestion before the reliance decision was made. Consider a chess player who has already identified a candidate move before seeing the AI suggestion. If the AI recommends the same move, the player enters the reliance decision in a state of prior agreement, which we define as alignment. Alignment is distinct from a reliance decision in that alignment does not always mean acceptance, as the user is still left with the decision to accept, reject or modify AI suggestion. It is also distinct from outcome agreement; two users can reach the same final move through different paths, only one of which involving prior alignment. Alignment captures only this pre-decision state, encoded as $\text{Align}_n \in \{0, 1\}$ in our new formulation.

Some Judge-Advisor System (JAS) literature show that agreement with external advice sometimes produce confidence inflation without corresponding accuracy gains [56, 57]. This suggests that alignment can amplify the effect of experience on confidence updating. The fast component incorporates this amplification:

$$F_{\text{fast},n}^k = \alpha_{\text{fast}}^k \left((1 + g_{\text{align}} \cdot \text{Align}_n) E_n - C_n^k \right)$$

where $\alpha_{\text{fast}}^k \in [0, 1]$ controls sensitivity to current trial evidence. When alignment occurs ($\text{Align}_n = 1$), the experience signal's pull on confidence is amplified by a factor $(1 + g_{\text{align}}^k)$, where g_{align}^k controls the amplification strength. In the present parameterization, we bound $g_{\text{align}} \in [0, 1]$. This bound is not a theoretical claim that stronger amplification is not possible, but rather a normalization choice that keeps the amplification factor $(1 + g_{\text{align}})$ on an interpretable scale. Furthermore, we found through robustness experiments with higher upper bounds up to 10 that even with broader bounds, most fitted values of g_{align} remain between $[0, 1]$.

5. RESULTS

We evaluate the reformulated model by fitting it to the same chess dataset described in Section 3.3. We first examine the sensitivity of the partial weight assignment parameter ρ , then report step-by-step model accuracy improvement, full model fit, and fitted parameters.

5.1. Sensitivity of ρ

Because modify trials are sparse in the dataset (fewer than 5% of all trials), the data provides limited information for estimating a stable ρ at the participant level. In a per-participant sweep, only 62 of 100 participants had enough modify trials for a best-fitting ρ to be estimable, and among these, 64.5% of the best-fitting values lay at the parameter boundaries (0 or 1). This boundary-heavy pattern suggests limited identifiability of ρ in the present dataset. Therefore, we evaluate the sensitivity of ρ with all modify data points.

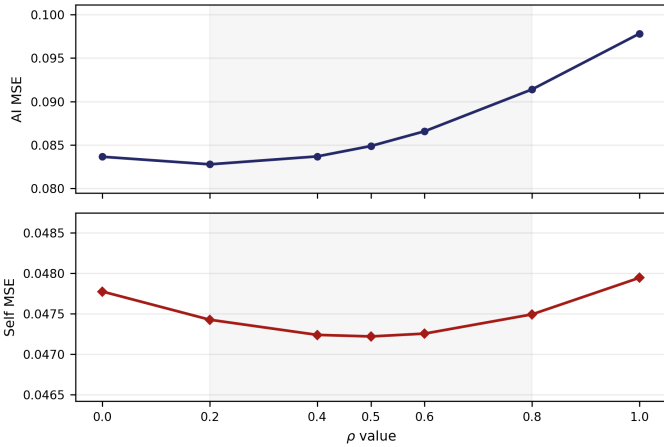


FIGURE 3: MSE of the model in modify trials as a function of the partial weight assignment parameter ρ for both AI and self channels.

Figure 3 shows the MSE of the model in modify trials as a function of $\rho \in [0, 1]$ for both channels. The MSE of the AI channel plateaus around $\rho \leq 0.5$ with the error increasing toward $\rho = 1$. The self-confidence channel shows a similar pattern at a smaller scale, with the minimum MSE values near $0.4 \leq \rho \leq 0.5$. In both channels, extreme values of ρ degrade the model fit. This pattern is consistent with individual participant fit results, where the mean best-fitting ρ is 0.519 ($SD = 0.438$). We therefore fix $\rho = 0.5$ in all subsequent analyses, treating modification as equal shared weight between accept and reject experience types.

5.2. Model Accuracy

Table 3 reports how model fit is impacted by each model change described in Section 4. The first stage traces the skeleton change from the original baseline to the physics-inspired form. The second stage tests each proposed change made to the physics-form baseline. Reported fitting results are means over 100 repeated random 80-participant subset refits, directly adopting the robustness test procedure conducted by Chong et al. [22]. Because some mechanisms operate primarily on sparse

or condition-specific subsets of trials, aggregate fit metrics alone may understate their contribution. We therefore supplement the fit comparison result with targeted analyses in the following subsections.

TABLE 3: Aggregate fit comparison

Model	C_{AI} MSE	C_{AI} adj. R^2	C_{self} MSE	C_{self} adj. R^2
<i>Skeleton Change</i>				
ORIG-BASE	1.60e-3	.753	1.35e-3	.424
PHYS-BASE	1.62e-3	.746	1.34e-3	.426
<i>Incremental Reformulation (applied to PHYS-BASE)</i>				
MOD	1.61e-3	.748	1.33e-3	.430
MOD+A	1.55e-3	.762	1.32e-3	.433
MOD+A+ASYM	1.55e-3	.762	1.32e-3	.433
MOD+A+ALIGN	1.34e-3	.792	1.15e-3	.508
PHYS-FULL	1.34e-3	.792	8.57e-4	.632

5.2.1. PHYS-BASE replaces the baseline’s additive error-correction to a physics-inspired form. The changes in the model accuracy are minimal (adjusted $R^2 = 0.424$ to 0.426 for self channel; adjusted $R^2 = 0.753$ to 0.746 for AI channel) as expected. The near-identical fit confirms that the skeleton swap does not distort model behavior.

With the chess dataset, the momentum parameter β^k converges near zero, so the skeleton reduces algebraically to the baseline’s single-step correction. Each trial is an independent chess puzzle with immediate feedback, and confidence is reported explicitly after every trial, leaving little temporal carry-over for momentum to capture. The near-zero estimate should therefore be interpreted as task-specific rather than as evidence against momentum effects in general. While PHYS-BASE does not improve the model accuracy on its own, the reformulated structure enables the subsequent changes without distorting the baseline model’s behavior.

5.2.2. MOD adds partial weight assignment for modify trials to the physics-based skeleton while keeping all other mechanisms unchanged. The improvement from PHYS-BASE is marginal (adjusted $R^2 = 0.426$ to 0.430 for self channel; adjusted $R^2 = 0.746$ to 0.748 for AI channel), which may be due to the following two reasons. First, modify encoding is a representational refinement that re-encodes existing trials rather than introducing a new driving signal. In addition, modify trials constitute less than 5% of the dataset, so even a more accurate encoding can have limited leverage on aggregate error.

Modification Effects. To further investigate how the modification encoding affects the model’s performance, we define a reliance decision-conditional fluctuation difference:

$$d(a) = \overline{|\Delta C^{AI}|_a} - \overline{|\Delta C^{self}|_a}$$

where $|\Delta C^{AI}|_a$ and $|\Delta C^{self}|_a$ are the mean absolute confidence changes for the AI and self channels, conditioned on reliance decision type $a \in \{\text{accept, modify, reject}\}$. This metric isolates each decision type and represents the systematic difference in the magnitude of confidence update between the two channels.

Figure 4 shows $d(a)$ with 95% confidence intervals for the empirical human data, the baseline model, and the full reformulated model. The results in this figure demonstrate that for accept and reject trials, the fluctuation difference between the two channels ($d(\text{accept})$ and $d(\text{reject})$) are estimated closely with the baseline model. However, for modification trials, the baseline does not capture this difference accurately. Experimental data show that $d(\text{modify}) > 0$, indicating larger confidence changes in the AI channel than in the self-confidence channel during modification trials. However, the baseline estimates the opposite with a negative $d(\text{modify})$ value. This confirms the need for a more accurate treatment of modification decisions in the baseline model, and the reformulated model improves the $d(\text{modify})$ value closer to the experimental data.

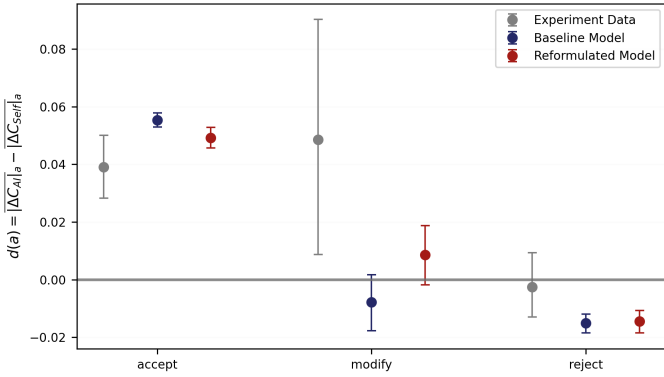


FIGURE 4: Baseline and reformulated models' reliance decision-conditional fluctuation differences $d(a)$ compared to experimental data

5.2.3. MOD+A redefines the accumulated confidence term as an accumulated expectation, so that A_n^k tracks experience rather than confidence history. The model accuracy is nearly unchanged from MOD for the self-confidence channel (adjusted $R^2 = 0.430$ to 0.433). This means that under symmetric coupling and without any alignment-differentiated experience, the two accumulator forms produce numerically similar predictions. The AI channel shows a modest improvement (adjusted $R^2 = 0.748$ to 0.762), suggesting that replacing a confidence-history accumulator with an experience-tracking expectation provides a slightly better reference for updating confidence in an external system. At this stage, this change is mostly representational, redefining the accumulator, but its payoff emerges in combination with the additional changes discussed below.

5.2.4. MOD+A+ASYM adds asymmetric slow learning between channels to MOD+A. The model performance is exactly identical to MOD+A (adjusted $R^2 = 0.762$ in the AI channel; adjusted $R^2 = 0.433$ in the self channel). Without alignment-differentiated signals, the accretive and error-corrective slow forms receive the same effective signals and therefore yield equivalent predictions.

5.2.5. MOD+A+ALIGN adds alignment amplification to MOD+A without asymmetric coupling. This produces the largest standalone improvement: adjusted R^2 rises from 0.762 to 0.792

($\Delta = +0.030$) in the AI channel and adjusted R^2 jumps from 0.433 to 0.508 ($\Delta = +0.075$) in the self channel. The disproportionate self-confidence gain suggests that, in this dataset, alignment events provide more predictive information for self-confidence than for confidence in AI.

Alignment Dynamics. Figure 5(a) shows the absolute prediction error $|C_{\text{experiment}} - C_{\text{Model}}|$ on alignment trials in both AI and self channels. With the reformulation, the prediction error on alignment trials is reduced in both channel (from 0.0696 to 0.0629 in the AI channel; from 0.0841 to 0.0554 in the self channel), with a greater reduction in self-confidence.

We examine this improvement further by defining the trial-level improvement as $|C_{\text{experiment}} - C_{\text{baseline}}| - |C_{\text{experiment}} - C_{\text{reformulated}}|$, so that positive values indicate trials where the reformulated model predicts more accurately than the baseline, and negative values indicate the opposite. Correlation results show that in the AI channel, higher magnitudes of improvement is associated with higher alignment case counts (Spearman $\rho = 0.402$, $p = 0.034$). However, in the self channel, this correlation is statistically insignificant (Spearman $\rho = 0.311$, $p = 0.107$). These results suggest that better prediction of self-confidence levels in the alignment trials is not the main driver of the large alignment-related gain in the self-confidence channel.

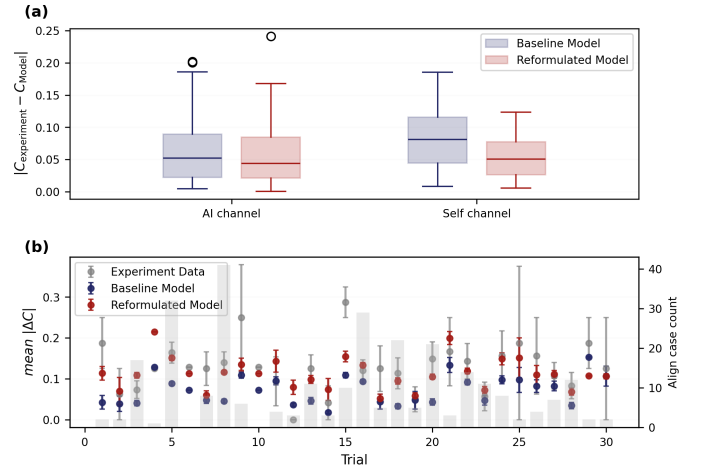


FIGURE 5: Model accuracy improvement under alignment events. (a) Absolute prediction error $|C_{\text{experiment}} - C_{\text{Model}}|$ on alignment trials for confidence in AI and self-confidence channels, (b) Trial-level mean absolute confidence change $|\Delta C|$ on alignment trials for model predictions, with shaded bars (right axis) indicating the number of alignment cases per trial.

The results in Fig. 5(b) offer a possible explanation. The panel shows trial-by-trial mean absolute confidence change $\Delta C = |C_n - C_{n-1}|$ on alignment trials based on both baseline and reformulated model predictions, with shaded bars indicating alignment case count per trial. The reformulated model tracks the empirical update magnitude more closely than the baseline, suggesting that alignment-related improvement may be associated not only with alignment case count, but also with the update magnitude. To test the hypothesis, we regress trial-level prediction improvement on empirical ΔC , alignment case count, and their interaction. In both channels, the interaction is positive and significant ($\beta = 0.055$,

$p = 1.41e - 10$ in the AI channel; $\beta = 0.130$, $p = 1.72e - 6$ in the self channel), confirming that the reformulated model performs better on alignment trials with larger empirical confidence changes. The interaction coefficient is roughly 2.4 times larger in the self-confidence channel, explaining the larger reduction in prediction error in this channel.

5.2.6. PHYS-FULL combines alignment amplification with asymmetric channel coupling. While adjusted R^2 in the AI channel remains at 0.792 just as MOD+A+ALIGN, adjusted R^2 in the self channel significantly increases from 0.508 to 0.632 ($\Delta = +0.124$). Figure 6 shows mean trial-by-trial confidence trajectories predicted by the baseline and the full reformulated model compared against the empirical data. In the AI channel (Fig. 6(a)), the two models produce similar mean trajectories, consistent with the modest aggregate improvement in the AI-channel fit. In the self channel (Fig. 6(b)), the reformulated model more closely tracks short-horizon deviations in the empirical trajectory, reflecting the larger improvement in self-confidence fit.

The gain from combining alignment and asymmetric slow learning is concentrated in self-confidence (+0.124), exceeding the standalone alignment gain (+0.075), even though asymmetric coupling alone does not produce any improvement. A plausible explanation is that, without alignment, the data do not contain sufficient variation to distinguish the accretive and error-corrective slow forms. Therefore, the two forms produce nearly identical fit (MOD+A+ASYM = MOD+A) because the optimizer can absorb their differences by adjusting other parameters. The addition of alignment changes this by amplifying the experience signal on alignment trials through the fast component, creating systematic shifts in the confidence trajectory. These shifts make the two slow learning forms more distinguishable in their predictions. The accretive form yields a better fit in the self-confidence channel than the error-corrective form, consistent with prior evidence that self-directed beliefs are less well described by symmetric error correction [51, 54].

5.3. Fitted Parameters

Table 4 lists the estimated optimal parameter values for the full reformulated model in each channel, from which two main patterns are notable. First, the alignment amplification coefficient g_{align} is substantially larger in the self-confidence channel with 0.70 compared to 0.17 in the confidence in AI channel, indicating that the fitted model assigns a larger amplification role to alignment in the self-confidence channel. Second, the expectation learning rate α_a^k is higher for confidence in AI with 0.083 than for self-confidence with 0.03, meaning the expectation state A_n^k adapts more quickly to new experience in the AI channel and drifts more slowly in the self channel.

TABLE 4: Representative fitted parameters for the full model.

	α_{fast}	α_{slow}	α_a	ω_1	ω_2	ω_3	ω_4	g_{align}
AI	0.23	0.52	0.083	0.84	0.19	≈ 0	0.57	0.17
Self	0.56	0.31	0.03	0.27	0.43	0.15	0.18	0.70

6. DISCUSSION

The reformulated model improves predictive accuracy across both confidence channels, with the largest gains in self-confidence (adjusted $R^2 = 0.424$ to 0.632). Beyond accuracy, the model’s representation of confidence dynamics in human–AI collaboration is better aligned with psychology, confidence in AI and self-confidence evolving separately through their own update mechanisms. The reformulation also improves the model’s applicability to human-AI design collaboration scenarios by capturing a broader range of interaction patterns.

6.1. Towards Confidence-aware Human-AI Interaction in Design

The reformulated model has implications for how AI-assisted design interfaces detect and respond to user confidence states. We highlight three aspects: source-specific diagnosis of miscalibration, the design risk posed by alignment amplification, and how these mechanisms may change in engineering design workflows.

6.1.1. Source-specific diagnosis. By separately estimating confidence in AI, self-confidence, and their update sensitivities, the model can identify both miscalibration and its likely source. Low reliance on AI may reflect justified distrust, inflated self-confidence, or both. Rapid acceptance of AI suggestions may reflect low self-confidence or agreement-sensitive over-updating. This distinction matters for intervention design. If the user appears overly confident in their own judgment but skeptical of the AI, the interface could provide stronger evidence for the AI recommendation, such as calibrated reliability cues, uncertainty information, or performance history [7, 9]. If the user appears inclined to trust the AI while lacking confidence in their own judgment, the interface could slow acceptance by requiring an initial independent response [18], side-by-side comparison, or brief justification before commitment [16]. Compared with a single global trust estimate, the dual-channel model supports more targeted calibration.

6.1.2. Alignment as a design-relevant signal. The fitted model suggests that alignment events disproportionately amplify self-confidence, with substantial variation across participants. Variance decomposition of alignment-trial prediction error supports this interpretation: in the AI channel, between-trial variance exceeds between-participant variance, and the pattern reverses for self-confidence, indicating that the self-confidence gain from alignment is dispersed across individuals rather than concentrated in specific trials.

One possible reason is that alignment is easier to interpret in the AI channel than in the self channel. For confidence in AI, agreement signals local compatibility between the user’s choice and the AI recommendation. For self-confidence, the same event may be read either as competence confirmation (“I identified a strong move”) or as validation of AI reliance (“AI guided me”) [4, 57, 58]. This ambiguity is consistent with the heterogeneity captured by g_{align} , although the present study does not test that mechanism directly.

Because the effect varies by person rather than by trial context, it cannot be managed simply by controlling when alignment information is presented. Different users will respond differently

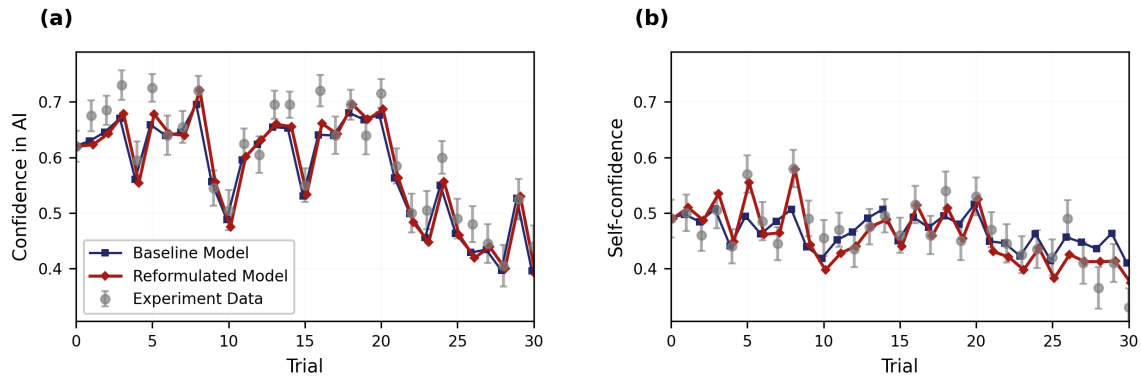


FIGURE 6: Mean confidence trajectories for confidence in AI and self-confidence

to the same agreement signal. For interface design, this raises a concern: explicitly signaling agreement between the user and the AI may increase self-confidence without improving decision accuracy [57]. However, if alignment sensitivity can be estimated from prior interactions, interfaces could adapt how agreement is presented. For users with high estimated g_{align} , agreement might be deemphasized, delayed, or paired with uncertainty information or alternative options. More broadly, convergence may be better framed as one cue among many rather than as validation [33].

6.1.3. Generalization to engineering design. The chess task shares common characteristics with AI-assisted engineering design tasks, allowing this reformulated model and the associated insights from this work to be applicable to various design scenarios. Both settings involve repeated evaluation of AI suggestions and reliance decisions. In both settings, users repeatedly evaluate AI suggestions, make reliance decisions, and form judgments about both the AI's and their own capability. Both scenarios also involve ambiguous credit assignment between human and AI contributions. These parallels make the chess task an appropriately controlled testbed in which the model's core mechanisms can be isolated and validated. Designers during an AI-assisted engineering design task are likely to display similar confidence mechanisms with the parameters adjusted for the specific task.

The partial weight assignment parameter ρ is especially salient in engineering design. In design scenarios, modification is likely the dominant interaction mode [30, 43], so confidence updates will need to accurately reflect human-AI joint contribution. Rather than doing an exhaustive sweep of ρ values, additional evidence can help separate those contributions to determine the appropriate ρ value. Prior design research has developed metrics for comparing geometric artifacts through geometric similarity, feature overlap, or constraint divergence [59, 60]. These measures may be utilized to quantify the degree of modification between an AI suggestion and the final design. In this way, the ρ values can be observable and carry greater explanatory value to confidence updates.

A similar extension applies to alignment. In chess, alignment appears as a discrete coincidence: the AI happens to recommend the move the user already identified. In design, convergence may emerge gradually across edits, distributing the amplification

effect across multiple edits. When AI adapts to the user through fine-tuning or personalization [61], this convergence becomes more frequent and meaningful. Under these conditions, the g_{align} component may be engaged more frequently, compounding self-confidence amplification in a way that reflects collaboration fit rather than competence.

Momentum (β^k) may likewise play a larger role in design. In the chess dataset, trials are independent and confidence is reported between trials, resulting in coarse, discrete updates that limit the continuity of confidence trajectories across trials. By contrast, engineering design often involves temporally extended interactions, where confidence develops over sequences of proposals, revisions, and evaluations. Under this conditions, momentum may become more prominent, making confidence update more path-dependent. In that sense, the timing of corrective intervention is more consequential.

6.2. Limitations and Future Work

Several limitations of the present study suggest directions for future work. First, the partial weight assignment parameter ρ for modify trials is fixed rather than fitted. This is appropriate for the present dataset, where the modify trials are sparse. However, in richer datasets, ρ may become more tractable by treating ρ as a standard global fitted parameter. Furthermore, because ρ is modification-specific, it may be better informed by artifact-level evidence of the extent of modification, as explained in the previous section.

Second, the experience signal assumes binary outcome feedback (positive/negative). In engineering design, outcome quality is typically evaluated along multiple dimensions, such as safety factor or maximum strain from FEA analysis. Feedback is therefore often presented as a performance score rather than a simple success-or-failure signal. Therefore, the discrete feedback and experience types could be replaced with a continuous encoding. More nuanced feedback can change how strongly people update their beliefs [44], providing useful insight for the model.

Third, the model assumes that all update parameters are stationary within a user. In longer design workflows, a user's sensitivity to feedback, alignment amplification, and momentum may change as the user becomes more familiar with the task.

Accommodating time-varying parameters would increase model flexibility, while potentially increasing the risk of overfitting. Therefore, a careful regularization or hierarchical estimation of time-varying parameters could improve the model further.

Finally, the interaction between the two confidence channels and its mapping to reliance decisions remains unexplored. The present model treats the two channels as coupled but distinct processes. Further work could examine whether specific patterns of co-evolution between confidence in AI and self-confidence correspond to their trust trajectories. Furthermore, the connection between this interaction and reliance decisions could be investigated, forming a more comprehensive framework useful for strategizing interventions.

7. CONCLUSION

This work proposes a reformulated dual-channel confidence model, predicting confidence in AI and self-confidence during AI-assisted design settings. From Chong et al.'s baseline model, the reformulation includes a new physics-inspired form that accounts for momentum of confidence change, an expanded interaction encoding with modification decision, a redefined accumulated expectation term, asymmetric slow learning across channels, and alignment amplification. Together, the reformulated model improves the prediction accuracy in both channels (AI and self), with the largest gain in self-confidence. The model's fit is adjusted $R^2 = 0.792$ in the AI channel and adjusted $R^2 = 0.632$ in the self channel. This model suggests a pathway from latent confidence dynamics to confidence detection and subsequent adaptive interventions in human-AI design workflows.

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During the preparation of this manuscript between February and March of 2026, the authors used ChatGPT (OpenAI, GPT-5 series) and Claude (Anthropic, Opus and Sonnet series) to identify about 10 of the cited references, debug data analysis scripts, and revise writing for clarity in Sections 2, 6, and 7. All outputs were reviewed and verified by the authors, and no AI-generated content was used directly without modification. The authors reviewed all edits and take full responsibility for all scientific content, analysis, and conclusions presented in this work.

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